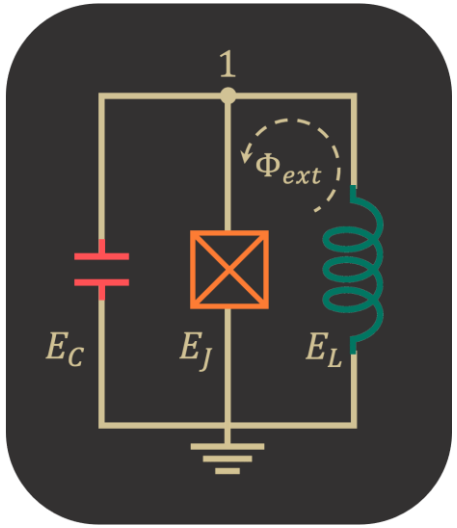
Fluxonium Chaos

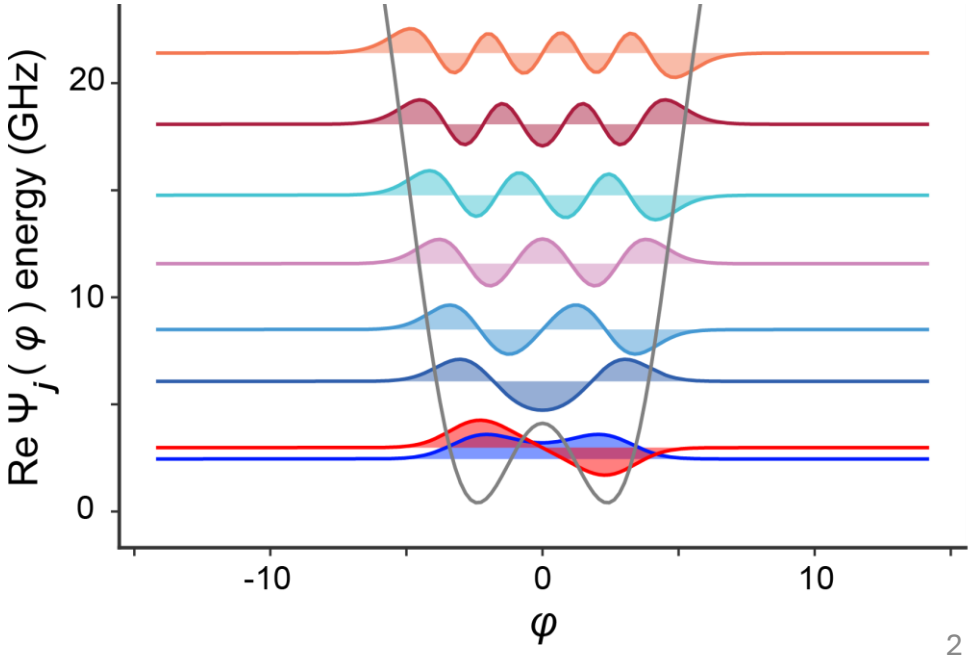
Static Hamiltonian



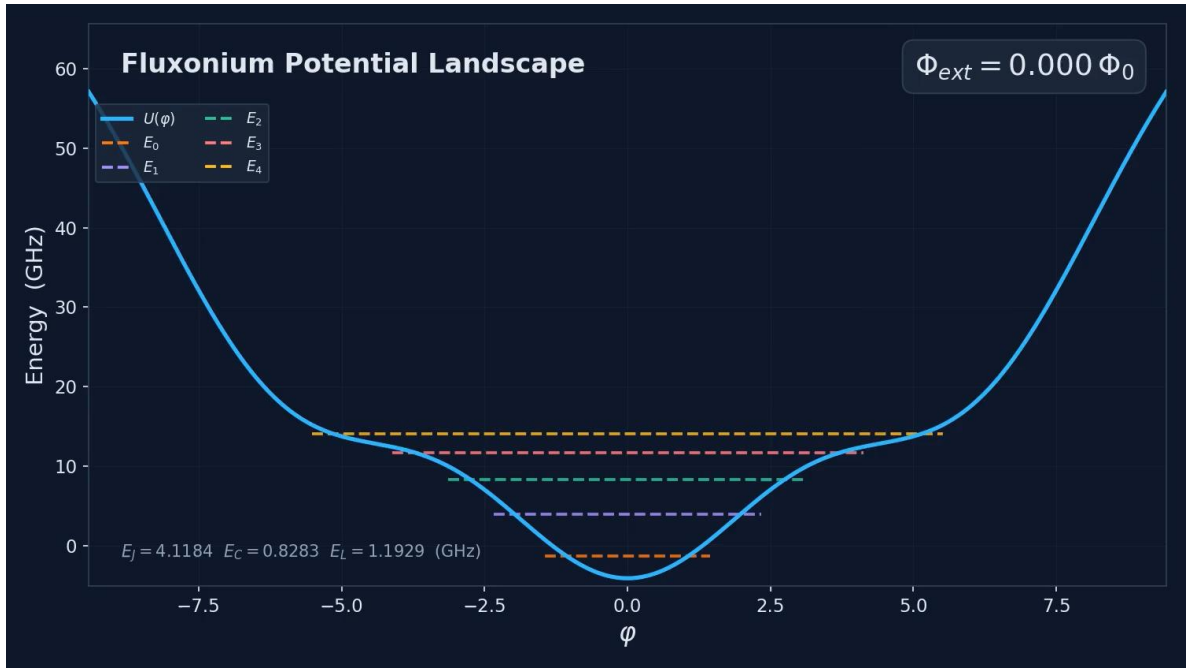
- $\hat{H} = 4E_C\hat{n}^2 + \frac{1}{2}E_L\hat{\phi}^2 - E_J\cos(\hat{\phi} - \phi_{ext})$

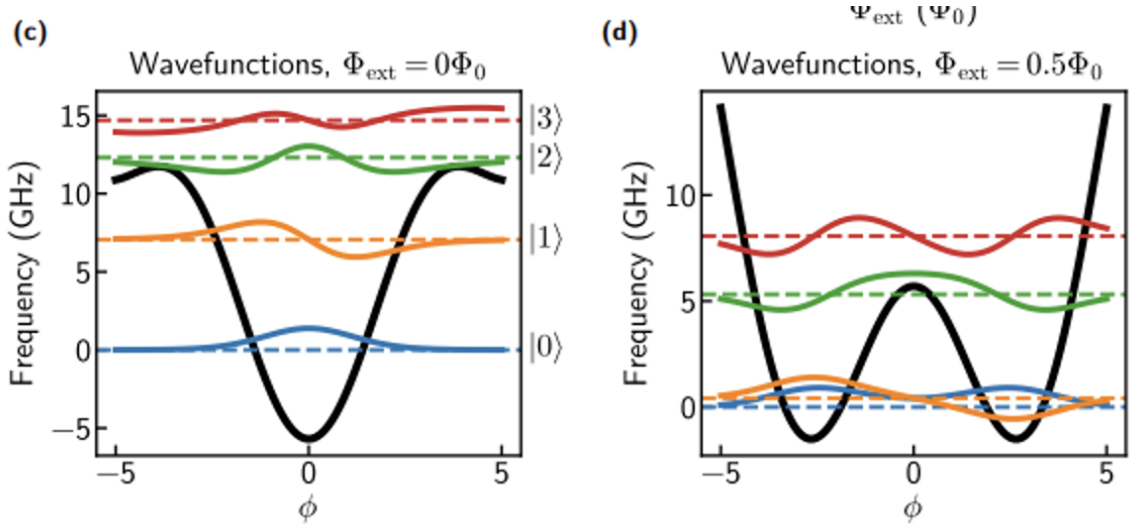
E_J = 4.1184 GHz
E_C = 0.8283 GHz
E_L = 1.1929 GHz
f_{01} = 525 MHz

f_r = 5.097 GHz
g = 28 MHz

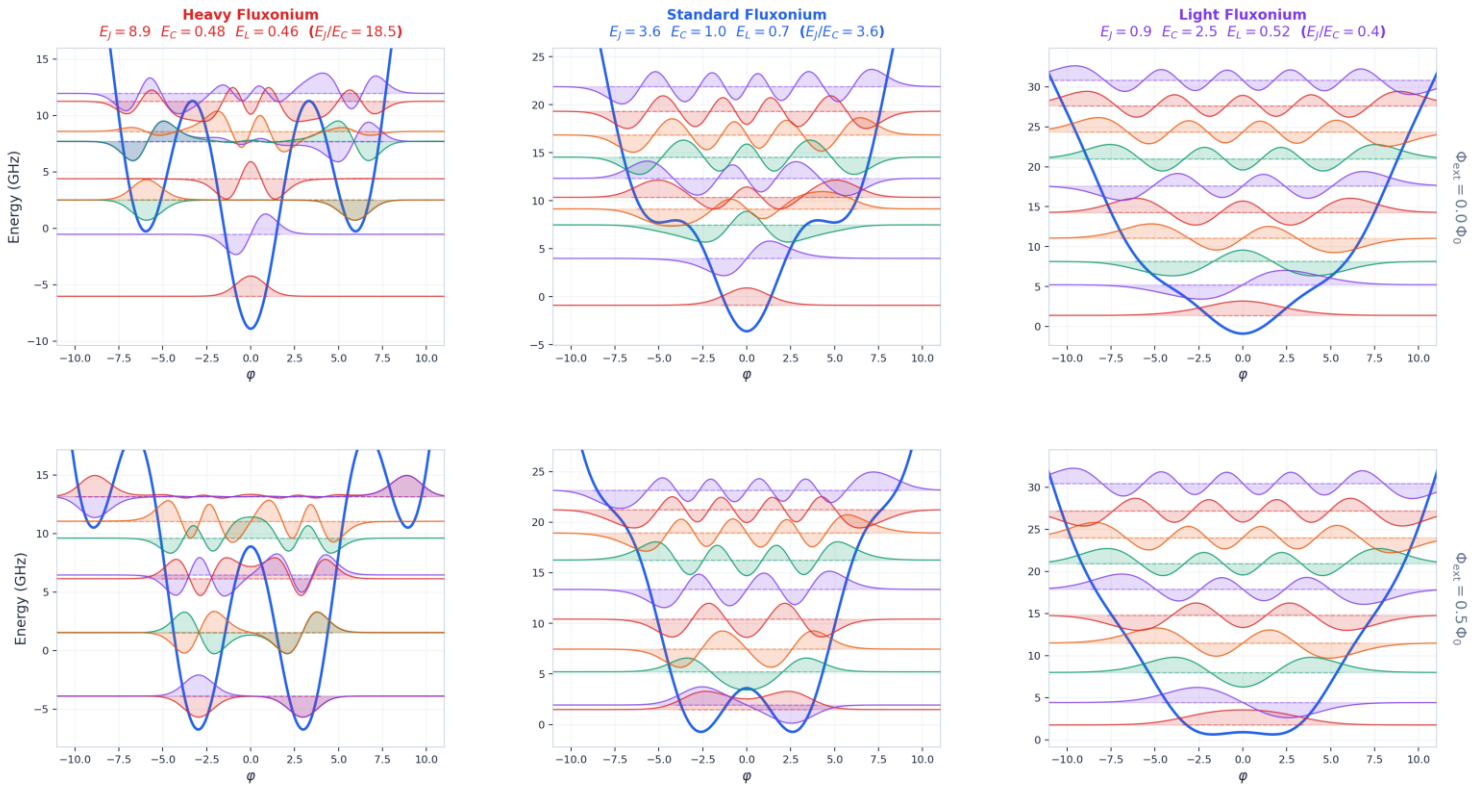


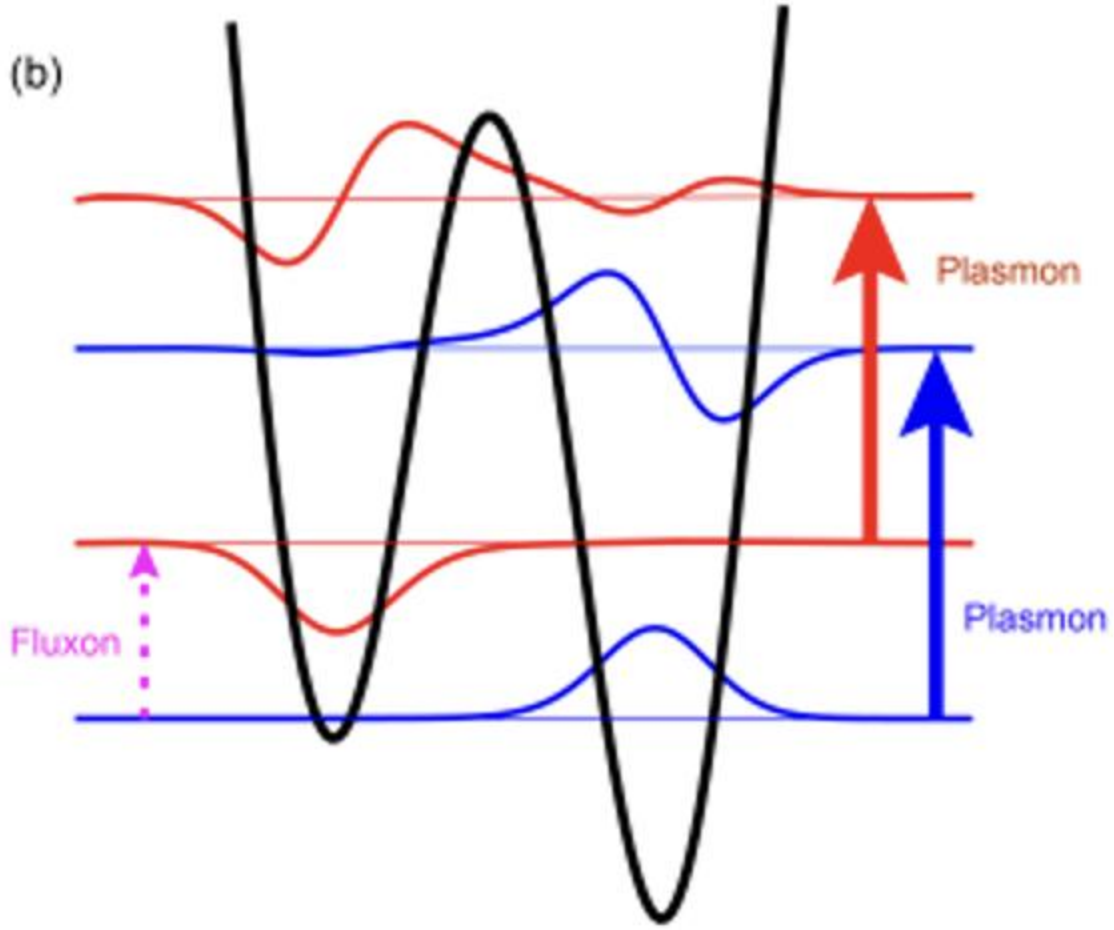
Potential Landscape – Tuning Across Flux





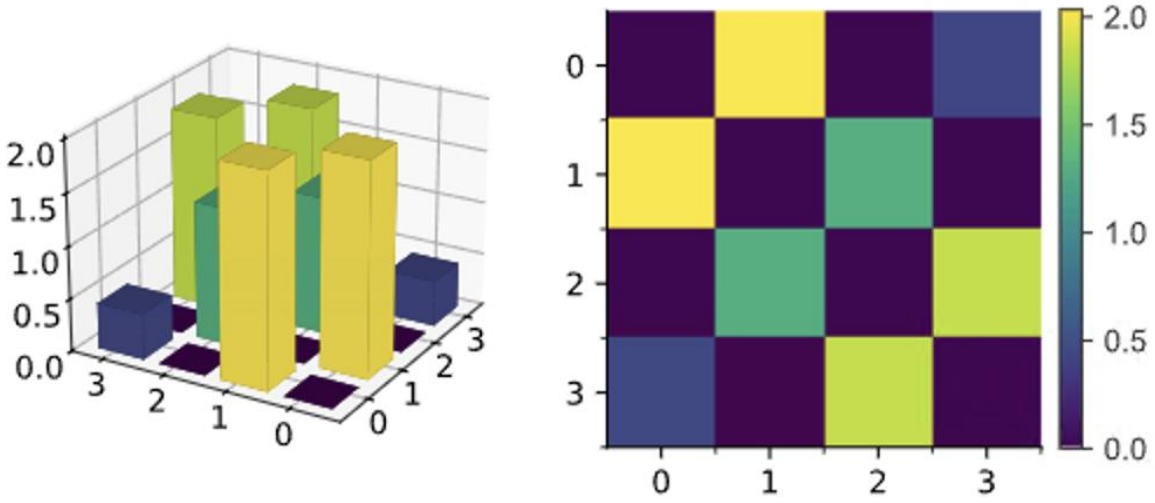
Potential Landscape – Regimes



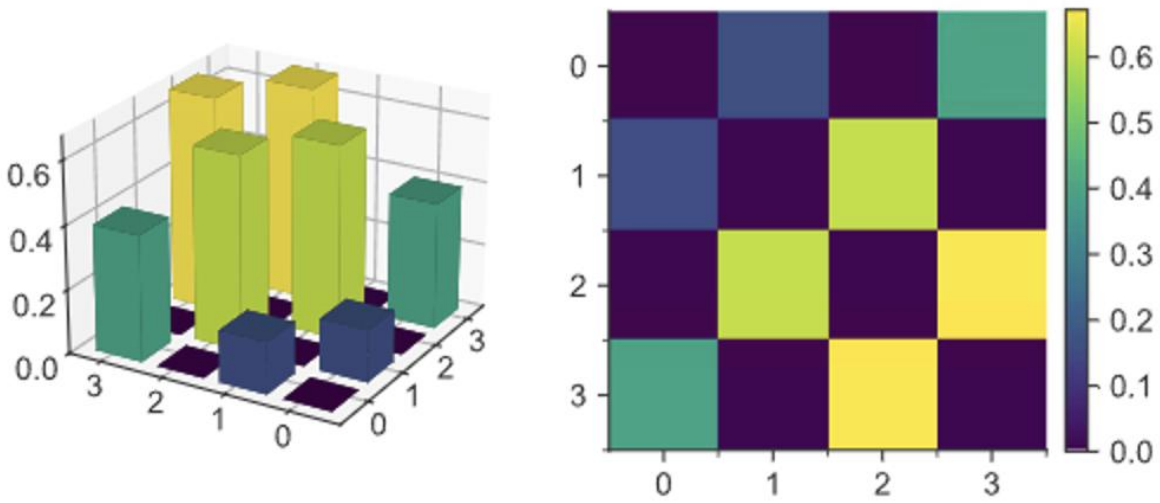

(b)
Plasmon
Plasmon
Fluxon

Matrix Elements

$\langle i|\hat{\phi}|j\rangle$



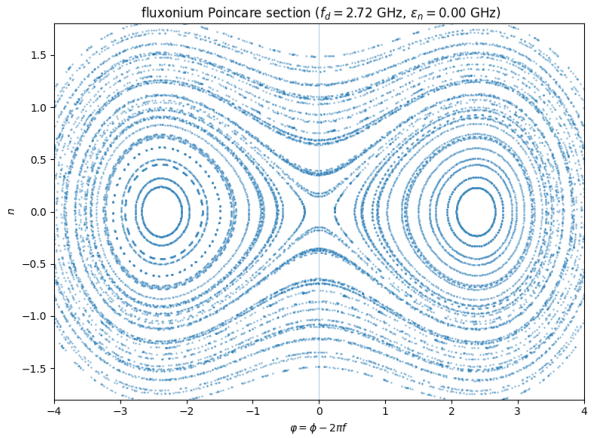
$\langle i|\hat{n}|j\rangle$

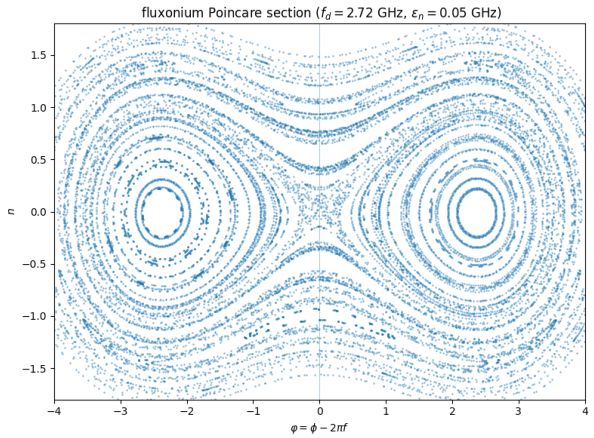


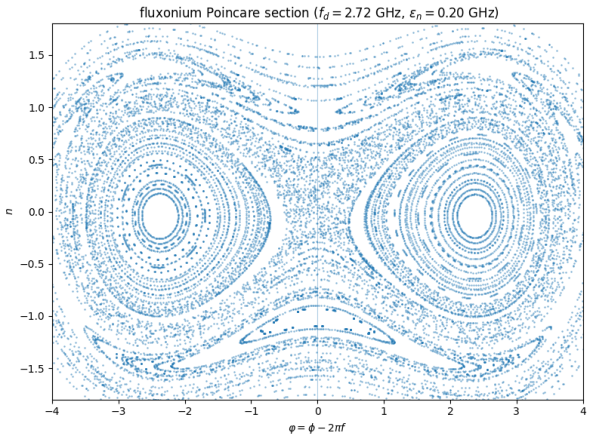
Classical Chaos

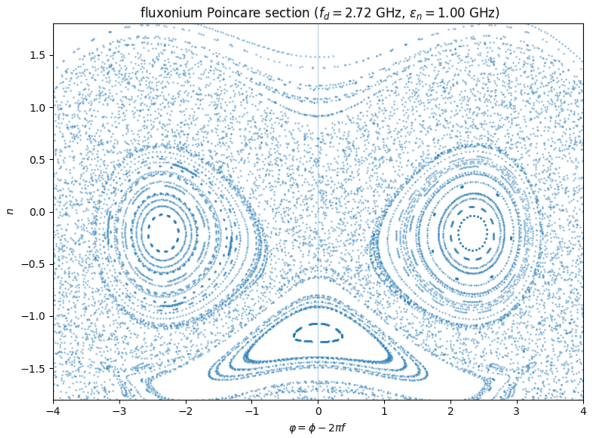
1. Poincare Sections
    1. Limited to just classical stroboscopic geometry
        1. Insights into where chaotic layers emerge
    2. Only intra-well motion
2. Lyapunov Exponents

Classical Chaos - Poincare



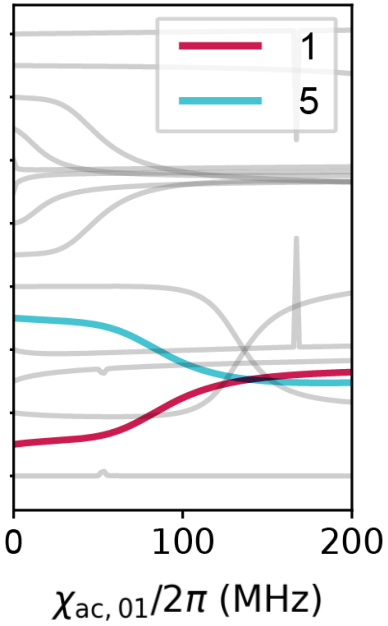






Quantum Chaos

1. Husimi distributions
2. Level statistics
3. Participation ratio and Shannon entropy



Allowed pathways open up as drive grows

↓

Multilevel hybridization of Floquet states

↓

Disjoint/delocalized Husimi support

↓

Stronger level repulsion + larger entropy and participation